

FORM TP 9196

JUNE 1991

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

MATHEMATICS

Paper 2 - General Proficiency

2 hours 40 minutes

1. Answer ALL the questions in Section I, and any TWO in Section II.
2. Begin the answer for each question on a new page.
3. Full marks may not be awarded unless full working or explanation is shown with the answer.
4. Mathematical tables, formulae and graph paper are provided.
5. Slide rules and silent electronic calculators may be used for this paper.
6. You are advised to use the first 10 minutes of the examination time to read through this paper and to plan your answers. Writing may begin during this 10-minute period.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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SECTION I

Answer ALL the questions in this section.

1. All steps in your calculation must be clearly shown.

(a) Calculate the exact value of:

$$\left(2\frac{1}{3} - 1\frac{5}{6}\right) \div 1\frac{1}{3}$$

(4 marks)

(b) The simple interest on \$15 000 for 4 years is \$8100.

Calculate the rate per cent per annum. (3 marks)

(c) The sum of \$2500 is divided among Peter, Queen and Raymond. Raymond received half, Peter received \$312.50 and Queen received the remainder.

Calculate (i) Raymond's share

(ii) Queen's share

(iii) the ratio in which the \$2500 was divided among the three persons

(iv) the percentage of the total that Peter received. (5 marks)

2. (a) Factorize completely:

(i) $1 - (a + b)^2$

(ii) $(2x^2 + xy - y^2) + 2x - y$. (3 marks)

(b) Solve:

$$\frac{x}{5} - 5 = \frac{x}{15} + 4$$

(4 marks)

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(c) Solve the simultaneous equations:

$$x + y = 5$$

$$xy = 6$$

(5 marks)

3. (a) The scores obtained by a class of ten students in a test were:

2, 3, 4, 4, 5, 6, 6, 7, 11, 12.

Calculate

(i) the median mark

(ii) the mean mark

(iii) the standard deviation of the marks; showing clearly all steps in your calculation. (8 marks)

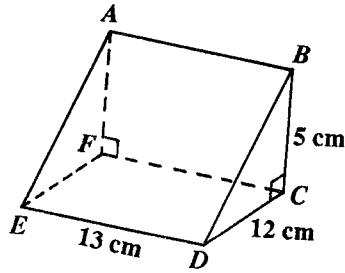
(b) Determine the probability that a pupil selected at random from the class obtained

(i) exactly 6 marks

(ii) at LEAST 6 marks. (4 marks)

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4. (a)



The figure $ABCDEF$ above, **not drawn to scale**, represents a wedge with measurements as shown. BC is perpendicular to the plane $FEDC$.

Calculate

- (i) the length, in cm, of BD
- (ii) the surface area, in cm^2 , of the wedge
- (iii) the volume, in cm^3 , of the wedge
- (iv) the size of angle BDC . (7 marks)

- (b) Using ruler and compasses only, construct a parallelogram $ABCD$, such that $AB = 6.5 \text{ cm}$, $AD = 5.7 \text{ cm}$ and the angle $DAB = 60^\circ$. Measure and state the length of BD in centimetres.

[Note: All construction lines must be clearly shown.]

(4 marks)

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5. There are 50 students in Form VI.

All students study Mathematics.

17 study Biology.

18 study Chemistry.

24 study Physics.

5 study Physics, Chemistry and Mathematics.

7 study Physics, Biology and Mathematics.

6 study Chemistry, Biology and Mathematics.

2 study all four subjects.

- (a) Draw a carefully labelled Venn diagram to represent the data, using the universal set as the set of students who study Mathematics. (3 marks)
- (b) Determine the number of students who study at LEAST TWO subjects. (2 marks)
- (c) Calculate the number of students who study Mathematics only. (1 mark)

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6. (a) The functions f and g are defined by:

$$f : x \rightarrow 5 + x$$

$$g : x \rightarrow x^2$$

Determine expressions for the functions:

(i) fg

(ii) g^{-1} (4 marks)

- (b) The distance S , in metres, moved by a particle from its starting point, at time t seconds, is given by

$$S = 3t + t^2$$

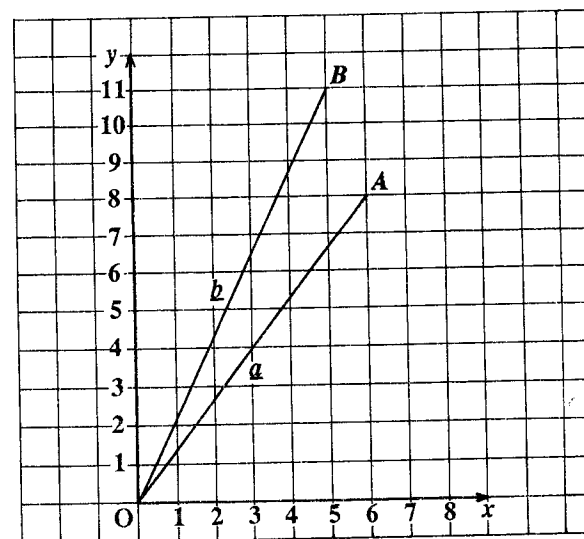
- (i) Copy and complete the table below for $S = 3t + t^2$.

t	0	1	2	3	4	5	6	7	8
S			10	18		40	54	70	

- (ii) Using a scale of 1 cm to represent 1 second and 1 cm to represent 10 m, draw the graph of the function $S = 3t + t^2$ for the range $0 \leq t \leq 8$.
- (iii) Using the graph, estimate the distance moved in 4.5 seconds.
- (iv) Draw a tangent to the curve at $t = 6$. Estimate the value of the tangent at this point. Give an interpretation of this value. (8 marks)

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7.



- (a) In the graph above, A and B are points such that $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$. The point P (not shown) is such that $\overrightarrow{OP} = \frac{1}{2}\underline{a} + \underline{b}$.

(i) Write $\overrightarrow{OP}$ in the form $\begin{pmatrix} x \\ y \end{pmatrix}$.

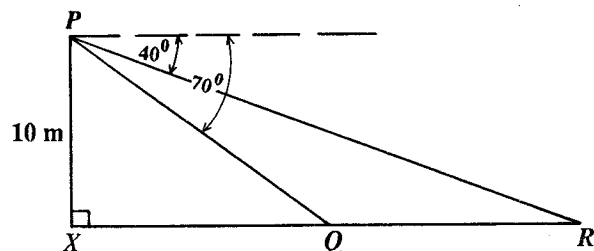
- (ii) Determine the length of OP . (5 marks)

- (b) The rectangle $WXYZ$, with coordinates $(2, 2)$, $(5, 2)$, $(5, 4)$ and $(2, 4)$ respectively, is mapped onto parallelogram $W'X'Y'Z'$ with coordinates $(5, 2)$, $(8, 2)$, $(10, 4)$ and $(7, 4)$ respectively under a shear S .

- (i) Plot the rectangle $WXYZ$ and its image $W'X'Y'Z'$.
- (ii) Determine the equation of the invariant line.
- (iii) Calculate the area of $W'X'Y'Z'$.
- (iv) State the coordinates of the image of the point $(2, 3)$ under the shear S . (7 marks)

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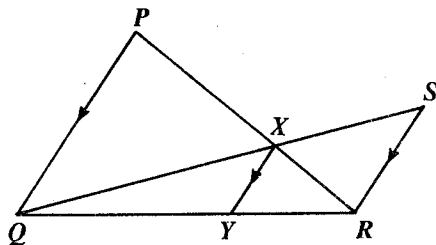
8. (a)



In the diagram above, not drawn to scale, the points X , Q and R are on a straight line in the same horizontal plane. The angle of depression of a point Q from the top of a tower PX , 10 m high, is 70° . The angle of depression of R from the top of the tower is 40° .

Calculate the length QR to one decimal place. (5 marks)

(b)



In the diagram above, not drawn to scale, PQ , XY and SR are parallel lines. $QY = 10$ cm, $YR = 5$ cm and $XY = 3$ cm.

- (i) Prove that triangles PQX and RSX are similar.
- (ii) Calculate the lengths of PQ and RS .
- (iii) Calculate the ratio of the areas of triangles PQX and RSX . (8 marks)

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SECTION II

Answer ANY TWO questions in this section.

TRIGONOMETRY

9. (a) Prove that $\frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} = \frac{1}{\sin^2 \theta \cos^2 \theta}$ (2 marks)
- (b) A ship sails 4 nautical miles from port P on a course 069° to a point Q , changes course to 295° and sails a further 5 nautical miles to a port R .
 - (i) Draw a carefully labelled diagram of the entire route taken. Show the north direction.
 - (ii) Calculate the distance PR to 2 significant figures.
 - (iii) Determine the bearing of P from R . (13 marks)
10. (a) If $\sin A = \frac{3}{5}$, $\cos B = \frac{15}{17}$ and A and B are acute angles, determine the exact value of $\tan A \tan B$. (3 marks)
- (b) [Take the radius of the earth to be 6400 km and π to be 3.142]
 - (i) The distance between Town P (80° E, 72° N) and Town R (x° E, 72° N) is 800 km. If Town R is east of Town P , determine x° E, the longitude of R . Give your answer to the nearest degree.
 - (ii) The shortest distance between Town P (80° E, 72° N) and Town Q (100° W, y° N) is 10054.4 km. Determine y° N, the latitude of Q . (12 marks)

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RELATIONS AND FUNCTIONS

11. (a) (i) Use the method of completing the square to determine the value of x for which the expression $4x^2 + 4x + 3$ is a minimum.
- (ii) Hence deduce the minimum value of $4x^2 + 4x + 3$.
(6 marks)
- (b) A motorist starting a car from rest, accelerates uniformly to a speed of 40 kmh^{-1} in 2 minutes. He maintains this speed for another 3 minutes. He then applies the brakes and decelerates uniformly to rest in 1 minute.
- (i) Draw a diagram of the velocity-time graph to show the different segments of the journey.
- Determine
- (ii) the acceleration, in kmh^{-2} , during the first two minutes
- (iii) the retardation, in kmh^{-2} , during the last minute
- (iv) the total length of the journey
- (v) the average speed for the whole journey in kmh^{-1} .
(9 marks)

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12. (a) Copy and complete the table below for $y = \frac{5}{x}$.

x	1	2	3	4	5	6
y	5.00				1.00	0.83

- (b) Using a scale of 2 cm to represent 1 unit on both x and y axes, draw the graph of the function above for $1 \leq x \leq 6$.
- (c) Using the same axes and scale given above, draw the graph of the function
 $x + y = 6$ for $0 \leq x \leq 6$.
- (d) (i) Shade on your graph the region which satisfies the inequalities
 $y \leq \frac{5}{x}$, $x + y \leq 6$, $x \geq 0$, $y \geq 0$.
- (ii) Indicate a point Z in the region which satisfies the inequalities
 $y \leq \frac{5}{x}$ and $x + y > 6$.
- (iii) Identify, by writing the letter T , the region bounded by the curve and the straight line.
- (iv) State the inequalities which describe the region bounded by the curve and the straight line.
(15 marks)

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VECTORS AND MATRICES

13. (a) If $A = \begin{pmatrix} 3 & 2 \\ 2 & -3 \end{pmatrix}$,

- (i) evaluate the determinant of A
- (ii) determine A^{-1}
- (iii) using A^{-1} solve the simultaneous equations:

$$3x + 2y = 1$$

$$2x - 3y = 5$$

(7 marks)

- (b) S is the transformation represented by

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

- (i) Perform the transformation S on a square with vertices at $A(0, 0)$, $B(0, 1)$, $C(1, 1)$, $D(1, 0)$ and write, in coordinate form, the images of A, B, C and D .
- (ii) Describe, in words, the single transformation represented by transformation S . (8 marks)

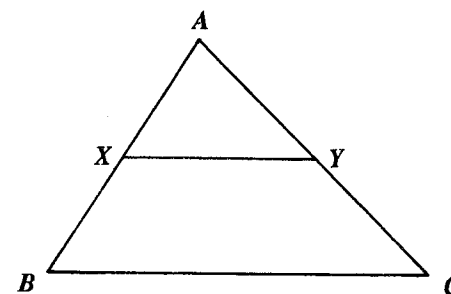
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14. (a) (i) T is the matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$.

Determine the elements of T which map $A(2, 1)$ onto $A'(4, 3)$ and $B(3, 2)$ onto $B'(6, 6)$.

- (ii) Derive the matrix S , such that TS represents the transformation matrix for a reflection in the x -axis. (10 marks)

(b)



The figure ABC is a triangle with X and Y the mid-points of AB and AC respectively. Using a vector method prove that

- (i) $\overline{BC}$ is parallel to $\overline{XY}$
- (ii) $BC = 2XY$. (5 marks)

END OF TEST